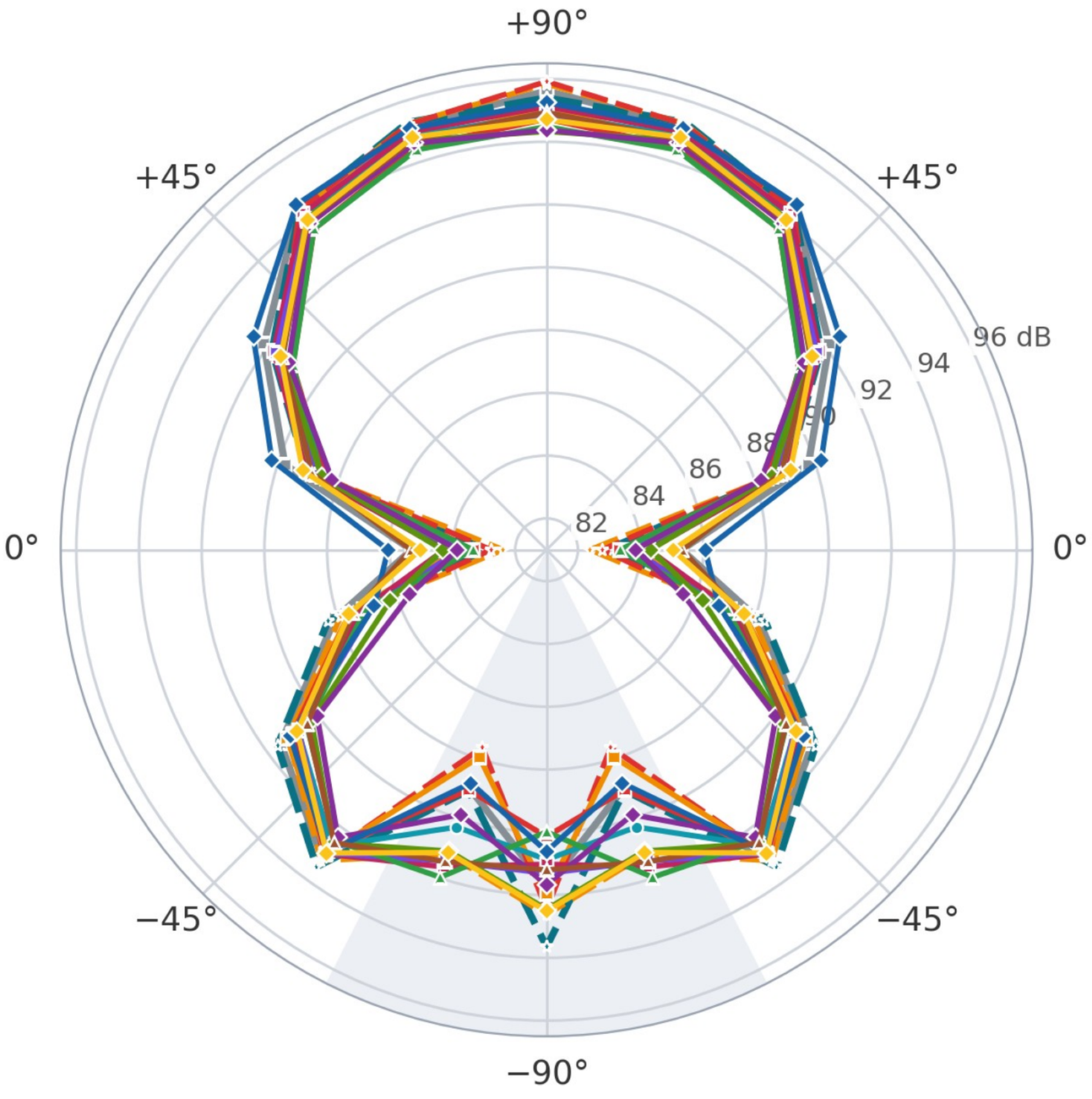


THE DUCT LININGS

Duct linings at 2000 µs

Mean level over the arc. Markers: ★ 3 Nov, ■ 2 Nov, ● 30 Oct, ◆ 29 Oct, ▲ 28 Oct. † lining fitted the old way (before 30 Oct). The bare prop is off the plot.

		level	thrust	rpm
Bare prop, no duct	■ 2 Nov	80.0 dB	**10.56 N**	11,807
Felt, double + rubber †	◆ 29 Oct	90.5 dB	**8.69 N**	12,177
Thinsulate, 4 layers	■ 2 Nov	90.6 dB	**9.22 N**	12,157
Felt †	▲ 28 Oct	90.7 dB	**8.96 N**	12,112
Inlet + Thinsulate, 4 layers	★ 3 Nov	90.9 dB	**10.05 N**	12,076
Thinsulate, 2 layers	■ 2 Nov	90.9 dB	**8.78 N**	12,205
Felt, double layer †	◆ 29 Oct	90.9 dB	**8.62 N**	12,196
Thinsulate, 6 layers	● 30 Oct	91.0 dB	**8.64 N**	12,176
Thinsulate, 10 layers	■ 2 Nov	91.0 dB	**8.89 N**	12,177
PU foam †	▲ 28 Oct	91.1 dB	**8.16 N**	12,234
Thinsulate, 8 layers	■ 2 Nov	91.2 dB	**8.82 N**	12,200
Inlet + Thinsulate, 2 layers	★ 3 Nov	91.3 dB	**10.04 N**	12,087
Rubber †	◆ 29 Oct	91.3 dB	**7.58 N**	12,292
PU foam + rubber †	◆ 29 Oct	91.3 dB	**8.33 N**	12,217
Duct + inlet, no lining	★ 3 Nov	91.4 dB	**10.12 N**	12,078
Duct, no lining	■ 2 Nov	91.5 dB	**7.99 N**	12,269

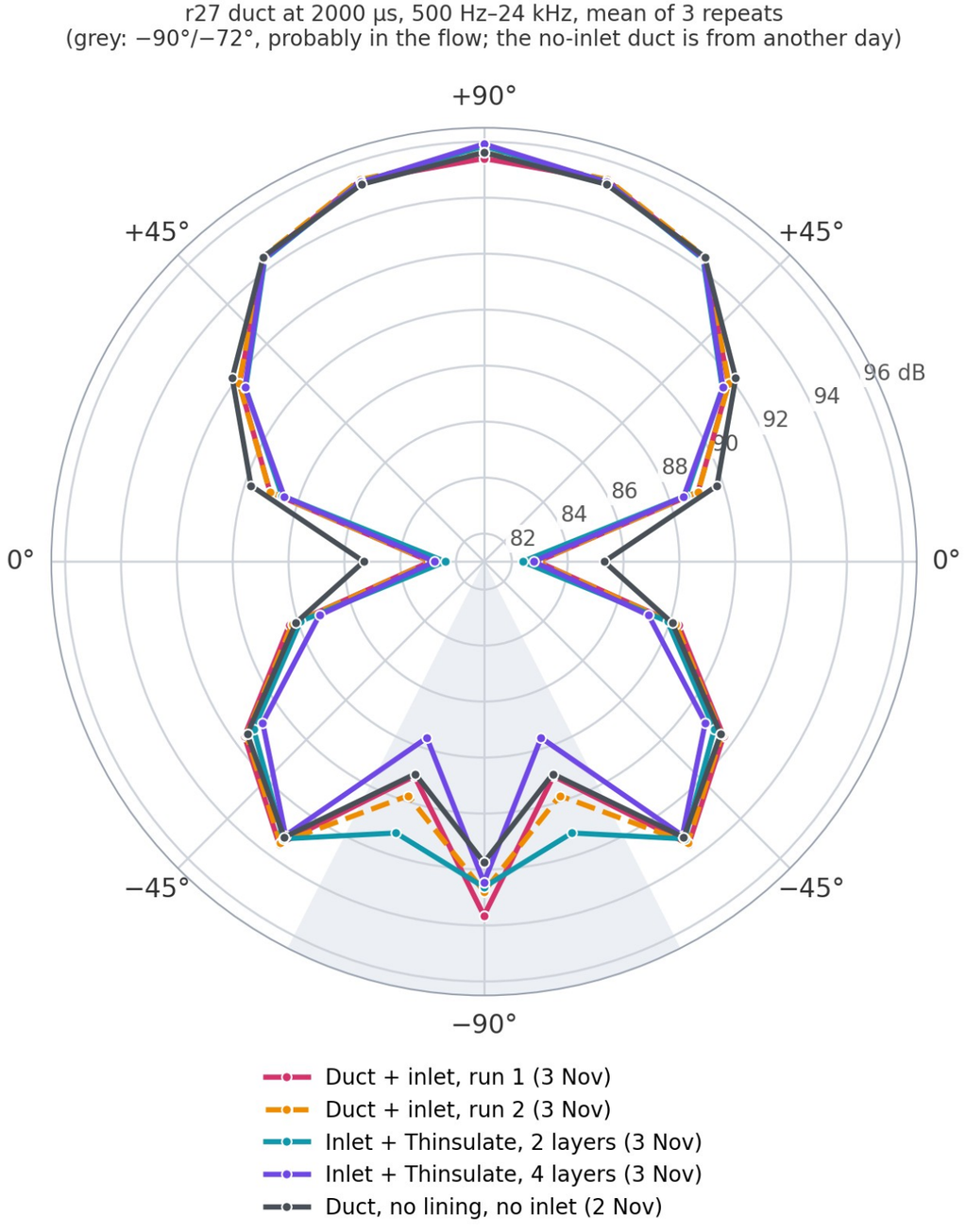
+1.23 N

Thinsulate, 4 layers against the unlined duct, both 2 Nov: it wins back 48 % of the 2.57 N the empty duct loses to the bare prop, and is 0.9 dB quieter from −54° to +90°.

Every duct is louder than the bare prop: 90.5–91.5 dB against 80.0 dB, at 10.56 N and 11,807 rpm bare. Set-ups marked † are from other days and the old fit: indicative only.

Inlet (dashed, 3 Nov): 10.12 N, +2.13 N over the unlined duct of 2 Nov at the same level (−0.3 dB from −54° to +90°). Same day with the inlet: Thinsulate 2 −0.08 N, −0.2 dB; Thinsulate 4 −0.07 N, −0.3 dB. A repeat of the bare inlet agreed to 0.04 N.

28 Oct–3 Nov 2026, APC 8x4.5MR, AXI 2217/12, 500 Hz–24 kHz, 2000 µs, mean of 3 repeats, corrected files. Axis 81–96.5 dB. Grey: −90°/−72°, probably in the flow. rpm = stand × 60/7.



THE INLET

With and without the inlet

2000 µs, mean of 3 repeats. Without the inlet: 2 Nov; with it: 3 Nov, so every row crosses days. Level is the mean over −54° to +90°.

lining	thrust no inlet	thrust inlet	Δ thrust	rpm no inlet → inlet	level no inlet → inlet	Δ level	0° no inlet → inlet
None	7.99 N	10.12 N	**+2.13 N**	12,269 → 12,078	91.7 → 91.4	**−0.3 dB**	85.3 → 82.9
Thinsulate 2	8.78 N	10.04 N	**+1.26 N**	12,205 → 12,087	91.1 → 91.2	**+0.0 dB**	83.9 → 82.4
Thinsulate 4	9.22 N	10.05 N	**+0.83 N**	12,157 → 12,076	90.8 → 91.1	**+0.3 dB**	83.3 → 82.8

Thrust: with the inlet all three land at 10.04–10.12 N. The weaker the set-up without it, the more the inlet adds, so the linings and the inlet fix the same loss. Per amp: 0.381 / 0.413 / 0.430 N/A without, 0.467 / 0.464 / 0.463 N/A with.

Noise: without the inlet the linings were 0.6–0.9 dB quieter than the empty duct; with it the three are within 0.3 dB, and the 0° microphone is quieter in every case. The inlet slows the prop by 80–190 rpm at the same ESC signal.

Caution: day to day the rig moves up to ~0.2 N and a few tenths of a dB, so the thrust gains are real and the level differences of 0.3 dB or less are not settled. Thinsulate 4 without the inlet, re-run on the inlet day, would make its row same-day.